

Product Requirements Document (PRD)

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SentinelAI

AI-Powered Autonomous Cyber Defense Twin

Version: 1.0 (Hackathon MVP)

Duration: 5 Days

Category: Cybersecurity • AI • Multi-Agent Systems • Knowledge Graph • Digital Twin

Executive Summary

SentinelAI is an AI-powered Cyber Defense Twin that creates a safe, isolated digital replica of an organization's infrastructure and autonomously simulates cyber attacks and defensive responses. Unlike conventional vulnerability scanners that merely list vulnerabilities, SentinelAI demonstrates how attackers exploit weaknesses, how defenders should respond, and why specific assets are at risk.

The platform combines a **Knowledge Graph**, **GraphRAG**, **Multi-Agent AI**, and an **Attack Simulation Engine** to provide explainable cybersecurity decision support. Instead of attacking real systems, SentinelAI operates entirely within a controlled lab environment using intentionally vulnerable applications, allowing organizations to visualize attack chains, evaluate defensive strategies, and prioritize remediation safely.

The MVP focuses on demonstrating autonomous attack planning, graph-based reasoning, AI-assisted defense recommendations, and executive-level reporting in an interactive dashboard.

Vision

Build an AI-powered cybersecurity platform that enables organizations to safely simulate attacks against a digital twin of their infrastructure, understand attack paths through graph reasoning, and automatically generate explainable mitigation strategies.

Mission

To transform cybersecurity from reactive vulnerability management into proactive cyber resilience by combining AI, graph intelligence, and safe attack simulation.

Problem Statement

Organizations face thousands of new vulnerabilities every year. Security teams typically rely on scanners that identify vulnerabilities but fail to explain:

- How attackers could chain vulnerabilities together.
- Which assets are truly at risk.
- Which vulnerability should be patched first.
- How effective existing defenses are.
- What business impact a successful attack could have.

As a result:

- Vulnerability reports become overwhelming.
 - Security teams spend hours manually correlating CVEs, network diagrams, and MITRE ATT&CK techniques.
 - Organizations often prioritize vulnerabilities incorrectly.
 - Executive stakeholders lack understandable risk insights.
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Solution Overview

SentinelAI creates a **Digital Twin** of an organization's infrastructure using a predefined infrastructure schema. It enriches this representation with software versions, known CVEs, MITRE ATT&CK mappings, and network relationships inside a Neo4j knowledge graph.

A Red Team AI agent analyzes the graph to determine the most likely attack path. Instead of executing real exploits, SentinelAI safely simulates each stage of the attack while collecting logs from an intentionally vulnerable lab application.

A Blue Team AI agent monitors these simulated attack events, recommends defensive actions, and estimates the reduction in organizational risk after each mitigation.

Finally, a Report Agent produces technical and executive summaries with a fully explainable "Why?" trace based directly on graph relationships.

Goals

Primary Goals

- Build a digital twin of infrastructure.
 - Simulate realistic cyber attack progression.
 - Use graph reasoning instead of keyword retrieval.
 - Demonstrate AI-powered defensive decision making.
 - Generate explainable security reports.
 - Visualize attack chains and mitigation effectiveness.
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Non Goals (MVP)

The following are intentionally excluded:

- Real malware execution
 - Live exploitation of external systems
 - Continuous network monitoring
 - Production SIEM integration
 - Live CVE synchronization
 - Cloud infrastructure integration
 - Automated penetration testing
 - SOAR automation
-

Target Users

Security Analysts

Need:

- Understand attack paths
 - Prioritize vulnerabilities
 - Generate reports quickly
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Security Operations Center (SOC)

Need:

- Analyze incidents

- Evaluate defensive readiness
 - Investigate attack progression
-

CISOs

Need:

- Executive risk summaries
 - Business impact analysis
 - Patch prioritization
-

Students & Security Researchers

Need:

- Safe cyber range
 - Learning platform
 - Attack visualization
-

User Journey

Step 1

Upload infrastructure

↓

Generate Digital Twin

Step 2

Knowledge graph created

↓

Software identified

↓

CVEs linked



MITRE mapped

Step 3

Red Team AI analyzes graph



Finds weakest entry point



Builds attack chain

Step 4

Attack simulation begins



Timeline updates



Logs generated



Dashboard updates

Step 5

Blue Team AI detects activity



Recommends mitigations



Calculates new risk

Step 6

Executive report generated



Explainability panel available

Functional Requirements

FR-1 Infrastructure Upload

Users upload a predefined JSON describing:

- Servers
- Applications
- Databases
- Firewalls
- Operating systems
- Network connectivity

Output:

Digital Twin

FR-2 Digital Twin Generator

Convert uploaded infrastructure into graph nodes.

Example

Server



Software



Operating System

↓

Connections

FR-3 Knowledge Graph

Store

- Assets
- Software
- CVEs
- MITRE ATT&CK Techniques
- Network Links
- Business Assets

Relationships

RUNS

CONNECTED_TO

HAS_CVE

EXPLOITS

MITIGATED_BY

CAN_REACH

FR-4 GraphRAG Engine

Retrieve graph neighborhoods instead of vector similarity.

Input:

Affected asset

Output:

Connected assets

CVEs

MITRE techniques

Reachable systems

Business assets

FR-5 Attack Simulation

Instead of executing exploits,

simulate

Reconnaissance

↓

Enumeration

↓

Initial Access

↓

Privilege Escalation

↓

Lateral Movement

↓

Objective Completion

Each step updates:

Timeline

Graph

Logs

Risk Score

FR-6 Red Team Agent

Responsibilities

- Entry point selection
- CVE selection
- Attack chain generation
- Business impact estimation

Output

Narrated attack plan

FR-7 Blue Team Agent

Responsibilities

- Detect attack events
 - Recommend mitigations
 - Prioritize patches
 - Estimate risk reduction
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FR-8 Risk Engine

Risk based on

- CVSS
- Exposure
- Reachability
- Asset criticality
- Patch availability
- Attack success probability

Output

Dynamic organizational risk score

FR-9 Explainability Engine

Every recommendation must include:

Why was this vulnerability selected?

Why is this attack possible?

Why is the risk score high?

Output generated directly from graph relationships.

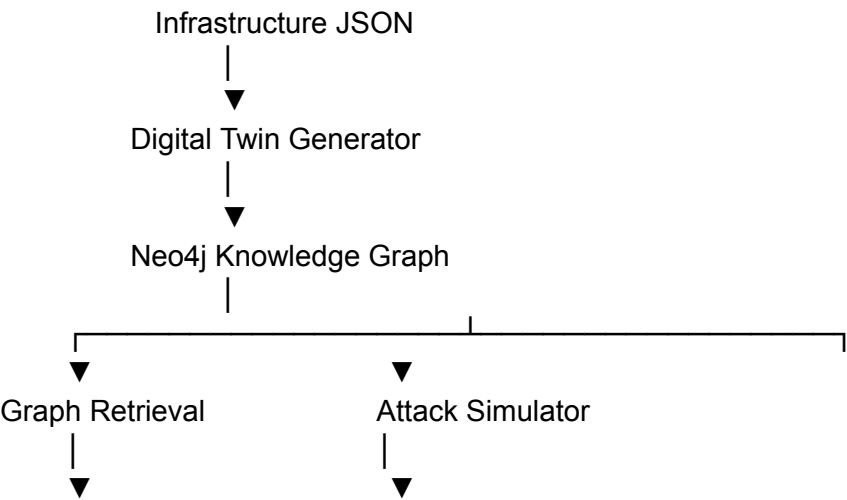
FR-10 Report Generator

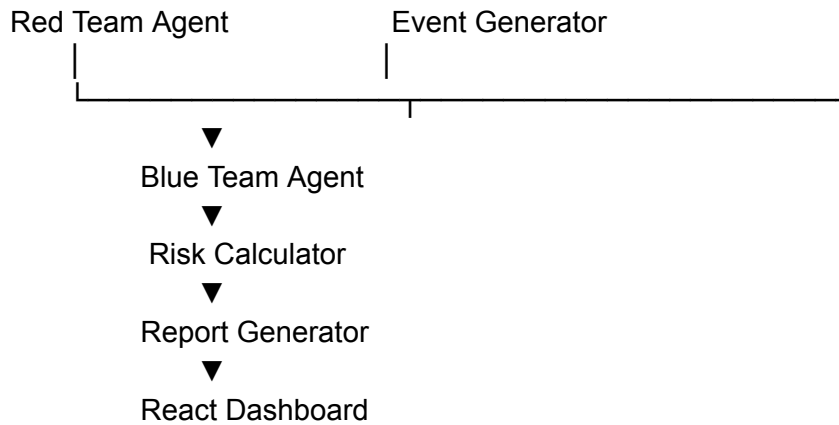
- Generate
 - Executive Report
 - Technical Report
 - Attack Timeline
 - Mitigation Plan
 - Business Impact
-

Non Functional Requirements

- Graph traversal < 2 seconds
 - AI response < 10 seconds
 - Dashboard updates in real time
 - Modular architecture
 - Explainable outputs
 - Safe simulation only
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System Architecture





AI Agent Design

Red Team Agent

Input

Infrastructure graph

Threat intelligence

Output

Attack chain

Entry point

Exploit sequence

Business impact

Blue Team Agent

Input

Attack events

Logs

Attack chain

Output

Mitigation recommendations

Risk reduction

Defense timeline

Report Agent

Input

Attack analysis

Defense analysis

Output

Executive report

Technical report

Explainability

Dashboard

Overview

- Overall Risk Score
 - Critical CVEs
 - Active Assets
 - Defense Effectiveness
-

Infrastructure Graph

Interactive Neo4j visualization

Attack Timeline

Live attack progression

Security Logs

Real-time simulated logs

AI Recommendations

Patch priority

Firewall updates

Network segmentation

Explainability

Graph reasoning path

Technology Stack

Layer	Technology
Frontend	React + Tailwind CSS
Backend	FastAPI
Graph Database	Neo4j AuraDB
Graph Algorithms	NetworkX + Neo4j Cypher
Agent Framework	LangGraph
LLM	GPT-5.5 / Gemini 2.5 Pro
Visualization	React Flow
Charts	Recharts
Dataset	Curated CVE + MITRE ATT&CK
Deployment	Vercel + Railway/Render

Demo Scenario

Scenario

A vulnerable Apache web server is connected to an internal database containing customer records.

SentinelAI performs the following sequence:

1. Loads the digital twin.
 2. Builds the knowledge graph.
 3. Identifies Apache as internet-facing.
 4. Maps Apache to a known CVE.
 5. Computes the attack path to the database.
 6. Simulates reconnaissance and exploitation steps.
 7. Streams realistic logs from the lab environment.
 8. Blue Team AI detects suspicious activity.
 9. Firewall rules and patches are recommended.
 10. Risk score decreases after mitigation.
 11. Executive report summarizes the incident.
 12. The "Why?" panel explains every decision through graph traversal.
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Success Metrics

Technical

- Infrastructure graph generated successfully.
- Attack path computed correctly.
- Graph traversal completes within target latency.
- AI agents produce coherent outputs.
- Dashboard updates in real time.

Demo

- End-to-end simulation completes within 3–5 minutes.
 - Explainability panel clearly justifies every recommendation.
 - Risk score visibly changes after defenses are applied.
 - Judges can follow the entire attack lifecycle without needing cybersecurity expertise.
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Future Roadmap

Phase 2

- Live CVE feeds (NVD/CISA)
- SIEM integration (Splunk, Elastic)
- Cloud asset discovery (AWS/Azure/GCP)
- SOAR integration
- Multi-user collaboration
- Attack scenario library
- Threat intelligence feeds

Phase 3

- Reinforcement learning for adaptive defense
 - Continuous attack simulations
 - Digital twin synchronization with production environments
 - Compliance reporting (NIST, ISO 27001)
 - Executive risk forecasting
 - Autonomous cyber defense recommendations
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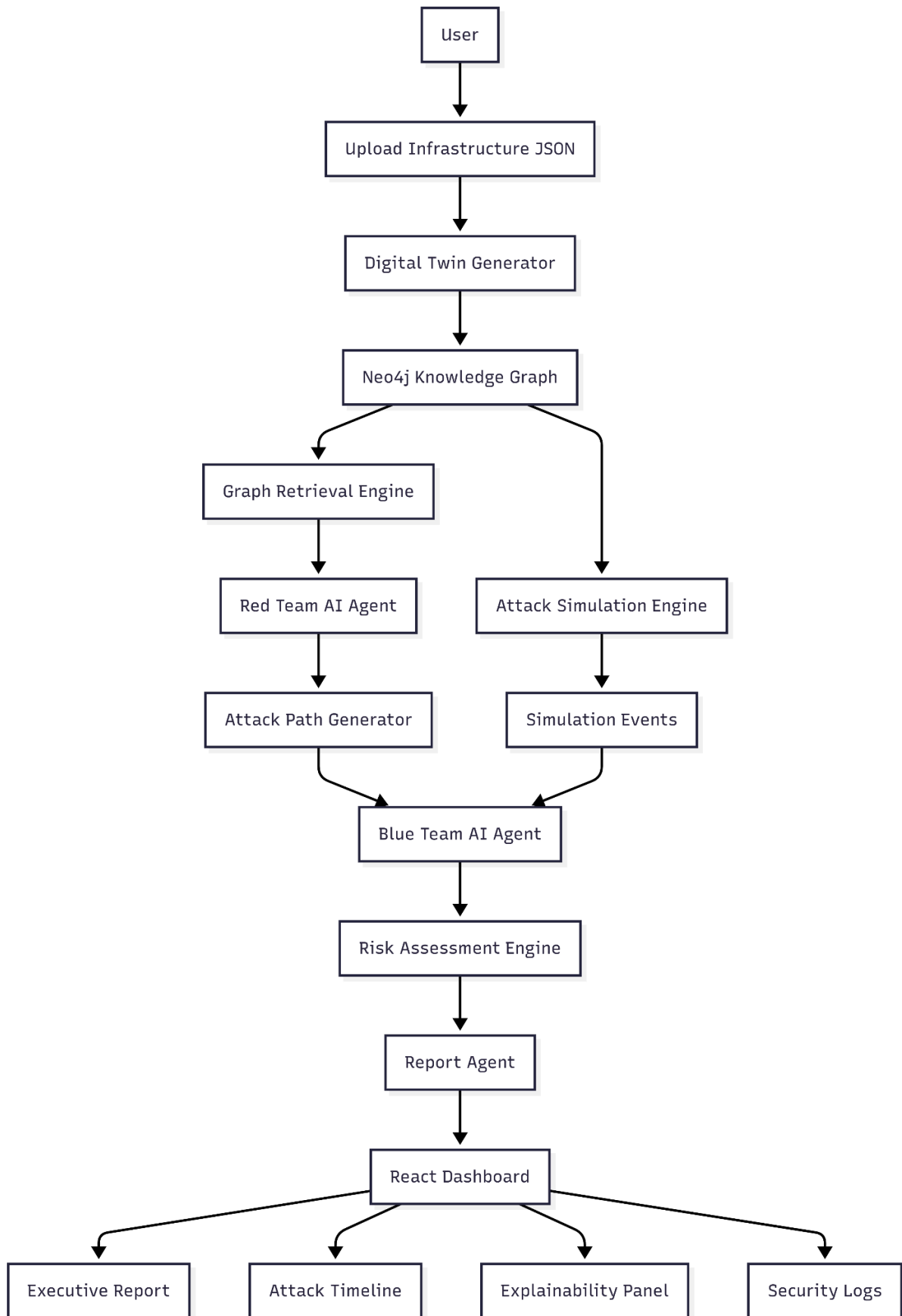
One-Line Product Pitch

SentinelAI is an AI-powered Cyber Defense Twin that safely simulates realistic cyber attacks on a digital representation of an organization's infrastructure, uses knowledge graph reasoning and multi-agent AI to explain attack paths, recommend defenses, and help security teams prioritize what matters most.

Proposed Project Architecture

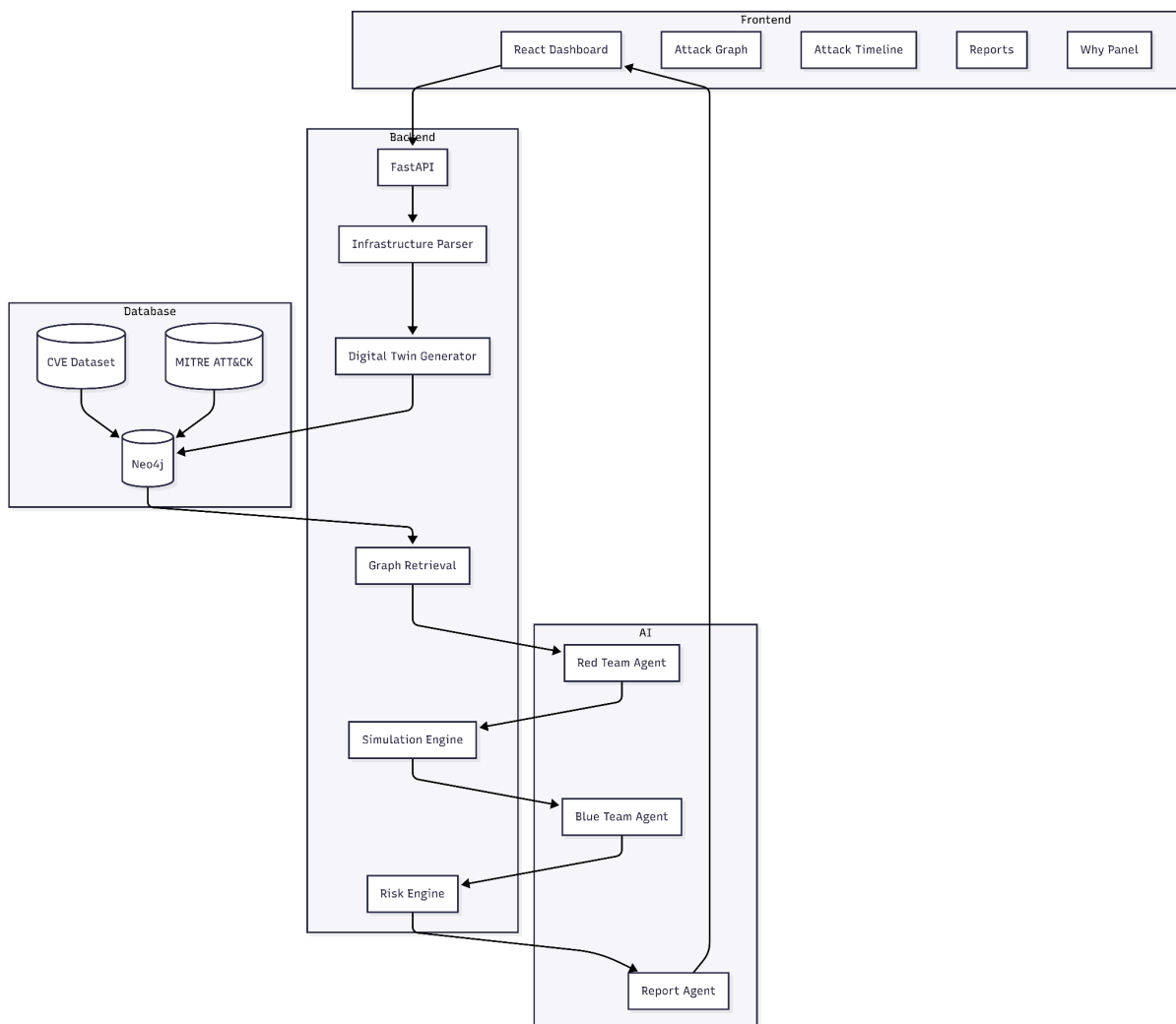
Proposed Project Architecture

Overall System Architecture



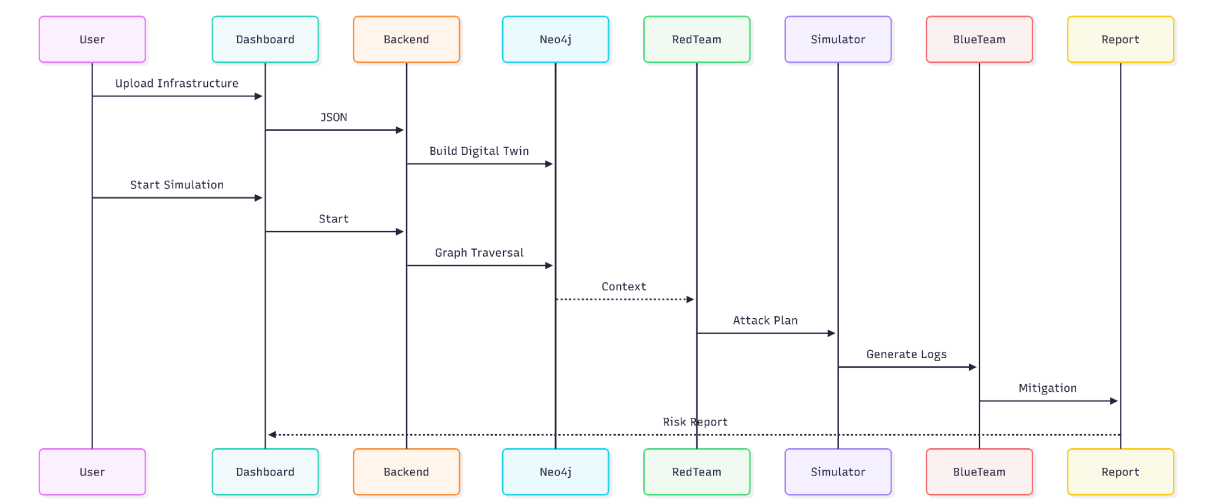
This diagram illustrates the complete architecture of SentinelAI and shows how data flows through the system. The process begins when a user uploads an infrastructure description, which is converted into a Digital Twin and stored as a Knowledge Graph in Neo4j. The graph is then queried using graph-based retrieval, allowing the Multi-Agent AI system to analyze potential attack paths, simulate cyber attacks, recommend defensive actions, calculate organizational risk, and generate explainable reports. Finally, all insights are presented through an interactive dashboard for security analysts.

Internal Component Architecture



This diagram explains the internal organization of SentinelAI by dividing the system into four major layers: Frontend, Backend, AI, and Database. The React frontend provides an interactive dashboard for visualization, while the FastAPI backend manages infrastructure parsing, graph construction, retrieval, and simulations. The AI layer contains specialized agents responsible for attack planning, defense recommendations, and report generation. The database layer stores infrastructure, CVE, and MITRE ATT&CK information in Neo4j, enabling efficient graph-based reasoning throughout the system.

Attack Simulation Workflow



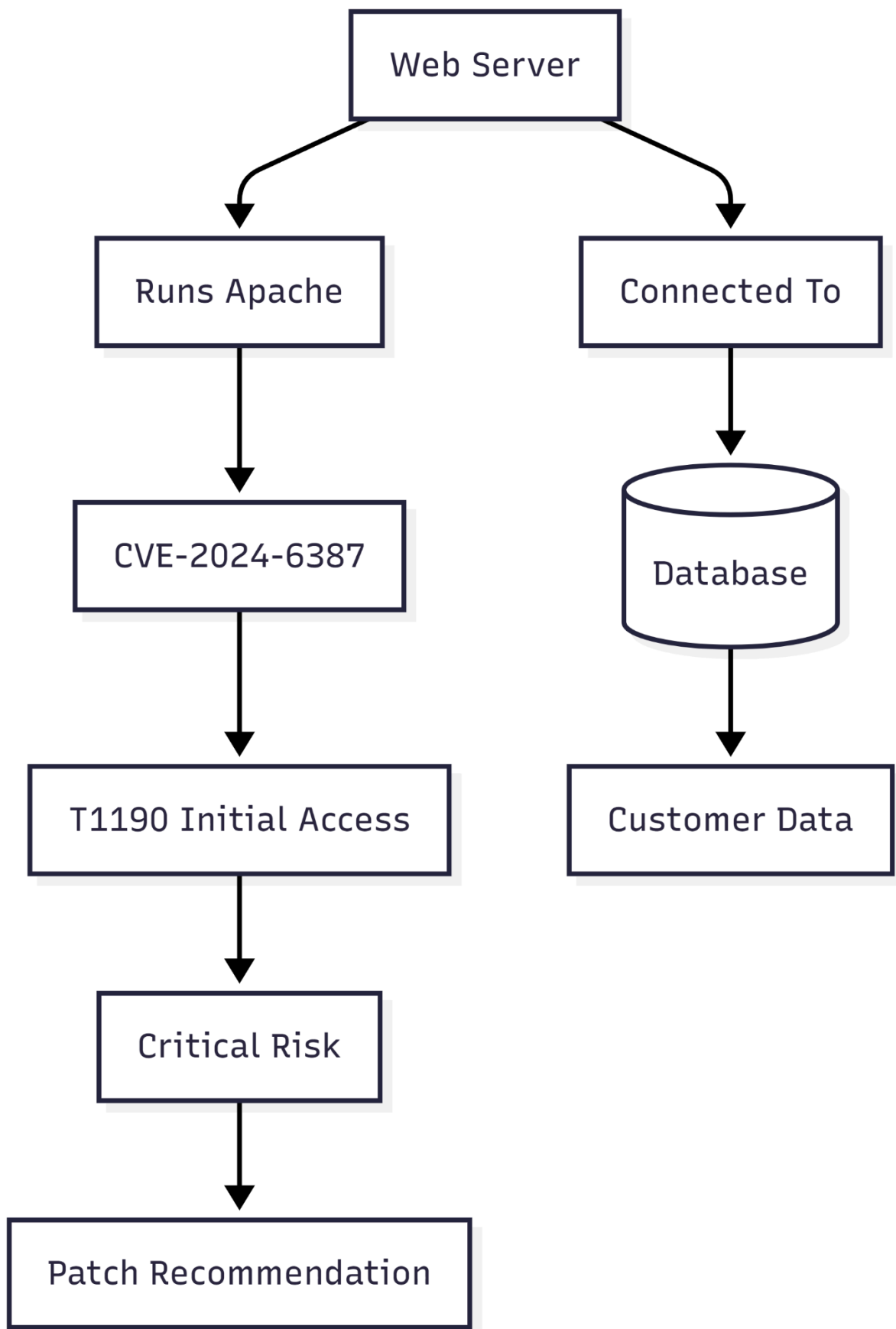
This workflow represents the complete execution sequence of a cyber attack simulation. After the user uploads the infrastructure and starts the simulation, the backend generates the Digital Twin and performs graph traversal to identify vulnerable attack paths. The Red Team Agent plans the attack based on graph relationships, the Simulation Engine generates attack events and logs, and the Blue Team Agent analyzes these events to recommend defensive actions. Finally, the Report Agent summarizes the attack, calculates the updated risk score, and presents the results through the dashboard.

Multi-Agent Interaction



This diagram demonstrates how multiple AI agents collaborate to automate cybersecurity analysis. The Red Team Agent first identifies the most likely attack path and vulnerable entry points. The Simulation Engine then executes a safe simulation of the attack and generates realistic security events. These events are analyzed by the Blue Team Agent, which recommends appropriate defensive actions. The Risk Assessment Agent recalculates the organization's security posture after mitigation, while the Report Agent generates both technical and executive-level reports with explainable reasoning for every decision.

Knowledge Graph Structure



This diagram represents the core intelligence layer of SentinelAI. Infrastructure assets, software, vulnerabilities, MITRE ATT&CK techniques, and business assets are modeled as interconnected nodes within a Neo4j Knowledge Graph. Rather than relying on traditional document search, the system traverses graph relationships to identify attack paths, understand asset dependencies, and explain why certain vulnerabilities pose higher risks. This graph-based representation enables more accurate, transparent, and explainable cybersecurity reasoning.

Tab 3

